

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Assessment Tool Weightage: 40%

QUESTIONS:

Total Marks:25

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Q5. Regularized CNN vs Non-Regularized CNN

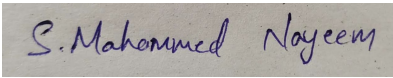
An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization** and a **CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

Code of Conduct:

I S.Mahammed Nayeem (192425294) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Q1. CNN vs Fully Connected Neural Network

A **Convolutional Neural Network (CNN)** is specifically designed to process grid-like data such as images, whereas a **traditional Fully Connected Neural Network (FCNN)** treats every input feature independently and connects it to every neuron in the next layer.

Aspect	CNN	Fully Connected Neural Network
Architecture	Convolution → Activation → Pooling → Fully connected/output	Fully connected layers throughout
Motivation	Designed to exploit spatial structure in images	Designed for general numerical/tabular features
Spatial feature extraction	Preserves and analyzes local spatial relationships	Spatial relationships are largely lost after flattening
Parameters	Relatively fewer because of parameter sharing	Very large number of parameters
Computational complexity	Lower for image processing	Very high for high-resolution images
Image handling	Naturally handles 2D/3D image structure	Usually converts image into a long 1D vector
Translation handling	Detects features even when their position changes	Less effective because each pixel has separate connections
Feature learning	Learns hierarchical features automatically	Learns features but without explicit spatial structure

Role of convolutional layers

The **convolutional layer** applies small filters, such as 3×3 or 5×5 , across the image. Instead of connecting every pixel to every neuron, it examines local regions.

For example:

Input X-ray → Convolution → Feature Map

A filter may learn to detect:

- Edges
- Lines
- Boundaries

- Textures
- Shapes

As the network becomes deeper, the learned features become more complex.

Role of filters

Filters are small matrices of learnable weights.

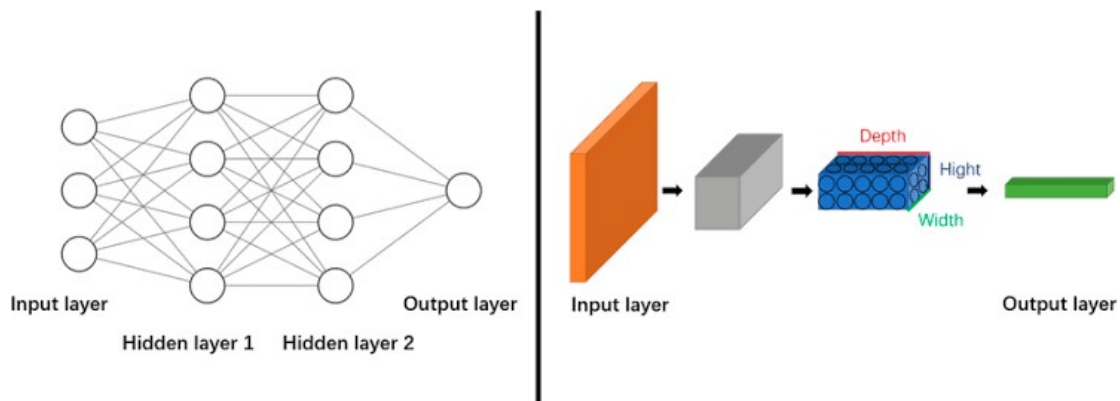
For example, an early filter might detect an edge:

$$\begin{bmatrix} -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

During training, the CNN automatically adjusts these weights to detect useful patterns in X-ray images.



Role of pooling

Pooling reduces the spatial dimensions of feature maps.

Common types:

- **Max pooling** – selects the maximum value.
- **Average pooling** – calculates the average value.

Pooling:

- Reduces computation
- Reduces memory requirements
- Helps retain important features

Why CNN is better for X-ray classification

An X-ray contains important spatial relationships—for example, the location and shape of abnormal regions matter. CNNs preserve these relationships while learning features automatically.

A fully connected network would require an enormous number of weights when processing a high-resolution X-ray.

Q2. Different CNN Layers and Filters

A CNN for industrial defect detection generally consists of:

Input → Convolution → Activation → Pooling → Fully Connected → Output

Comparison of CNN layers

Layer	Main function	Feature extraction	Spatial information	Computation
Convolution	Applies filters to input	Extracts local features	Preserves spatial arrangement	Moderate
Activation	Introduces non-linearity	Helps learn complex patterns	Preserves feature-map structure	Low
Pooling	Reduces dimensions	Retains important features	Reduces spatial resolution	Low
Fully Connected	Performs final classification	Combines learned features	Spatial structure mostly removed	High

1. Convolutional layer

The convolutional layer scans the product image using filters.

It can detect:

- Cracks
- Scratches
- Holes
- Broken edges
- Surface irregularities
- Texture variations

The output is called a **feature map**.

2. Activation layer

A common activation function is **ReLU**:

$$\text{ReLU}(x) = \max(0, x)$$

It converts negative values to zero and allows the network to learn non-linear relationships.

Without activation functions, stacking multiple convolutional layers would behave approximately like a single linear transformation.

3. Pooling layer

Pooling decreases the size of feature maps.

For example:

4×4 feature map

↓

2×2 Max Pooling

↓

2×2 feature map

This reduces computation while retaining strong responses corresponding to important defects.

4. Fully connected layer

The final learned features are combined by fully connected layers.

The network can then classify the product as:

Normal / Defective

or:

Scratch / Crack / Dent / Hole / Normal

Q3. Parameter Sharing vs Independent Weights

Basic idea

In a fully connected network, different connections generally have different weights.

In a CNN, the same convolutional filter weights are reused across different locations of an image. This is called parameter sharing.

Comparison

Aspect	CNN with Parameter Sharing	Fully Connected Network
Weights	Same filter reused spatially	Separate weights for connections
Number of parameters	Much smaller	Very large
Memory	Lower	Higher
Computation	More efficient for images	Expensive for high-resolution images
Feature detection	Same feature detected anywhere	Location-specific learning
Translation handling	Better	Poorer
Learning efficiency	High	Lower for image patterns
Scalability	Suitable for large images/datasets	Difficult for high-resolution images

Example

Suppose an image has $1,000 \times 1,000$ pixels.

That gives:

1,000,000 input values

If a fully connected layer has 1,000 neurons:

Parameters $\approx 1,000,000 \times 1,000 = 1$ billion weights

This creates huge memory and computational requirements.

Now consider a CNN with a 3×3 filter.

Only:

$3 \times 3 = 9$ weights

are required for one filter, plus a bias.

Those same 9 weights can be applied throughout the image.

Why parameter sharing is important

Suppose a CNN learns a filter for detecting a particular edge.

That filter can detect the same edge:

- At the top of the image
- At the bottom
- On the left
- On the right
- At different locations

Therefore, the CNN does not need to learn a separate edge detector for every pixel location.

Translation-related feature detection

Parameter sharing makes CNNs equivariant to translations at the convolution stage: when an object moves, the corresponding feature response also moves.

Pooling and other architectural choices can then provide some degree of translation invariance, making classification less sensitive to small changes in object location.

Conclusion

Parameter sharing dramatically reduces:

- Number of parameters
- Memory usage
- Training requirements
- Computational cost

Therefore, it is essential for processing thousands of high-resolution surveillance images efficiently.

Early-layer filters vs deeper-layer filters

Early layers	Deeper layers
Detect simple features	Detect complex features
Edges	Shapes
Lines	Object parts
Corners	Defect patterns
Simple textures	Complete defect structures
Small receptive fields	Larger effective receptive fields

For example:

Layer 1: detects edges
↓
Layer 2: detects curves and textures
↓
Layer 3: detects shapes and surface patterns
↓
Deep layers: recognize a complete crack or defect

How the layers improve defect detection

The layers work hierarchically:

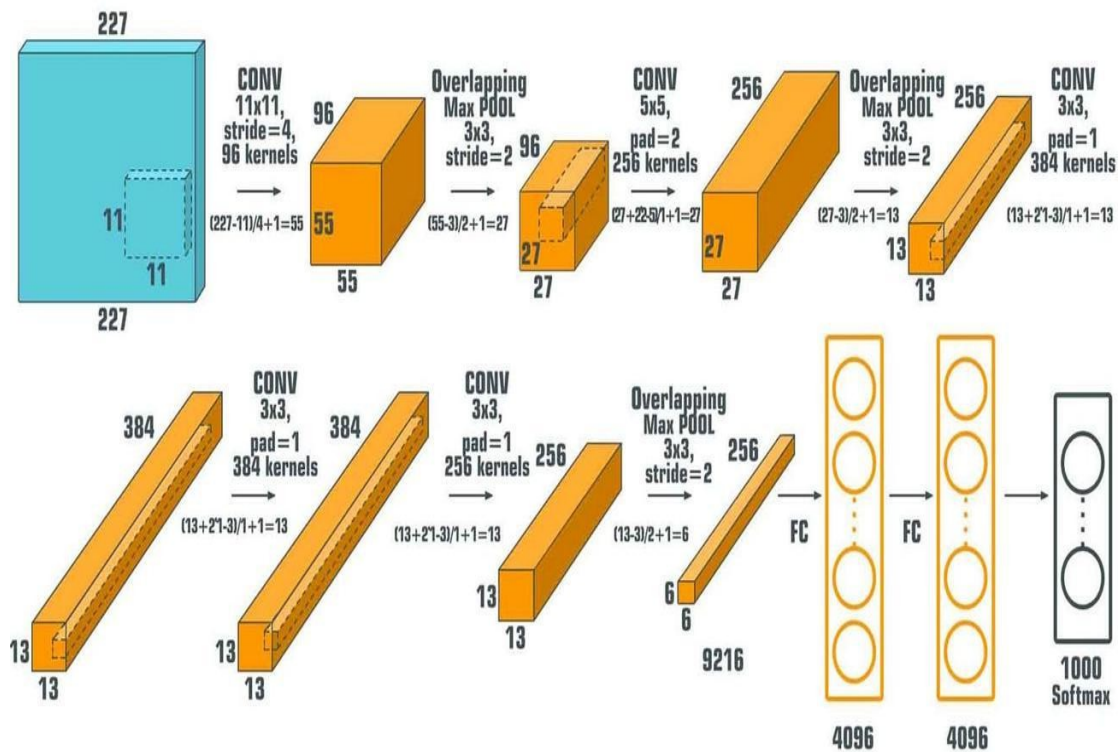
Raw image → edges → textures → shapes → defect patterns → defect class

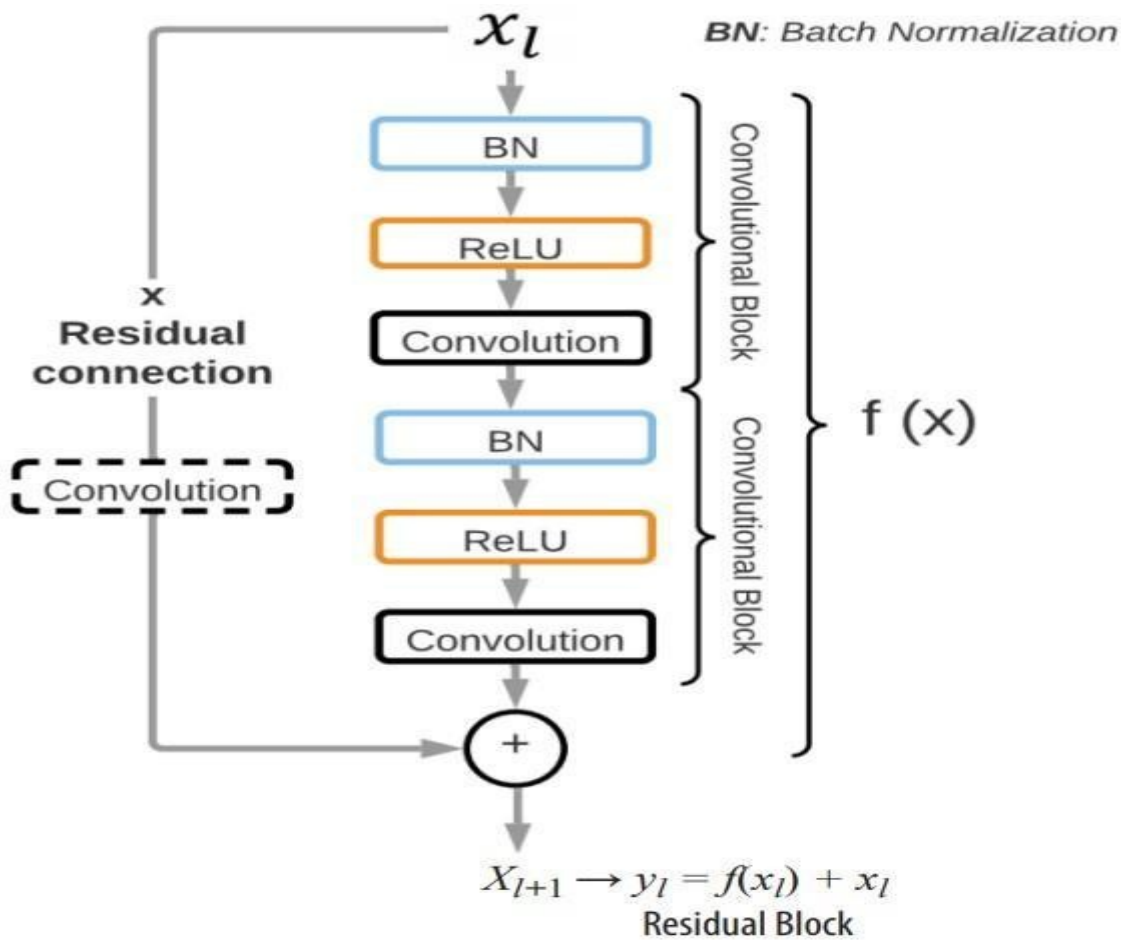
Thus, CNN layers allow the system to detect both simple and complex defects automatically.

Q4. AlexNet vs ResNet

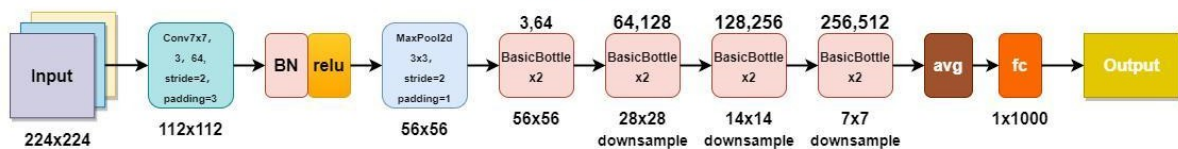
AlexNet

CONVOLUTIONAL NEURAL NETWORK

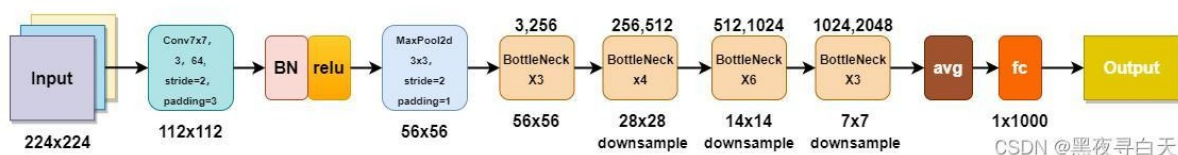




ResNet-18



ResNet-50



CSDN @黑夜寻白天

AlexNet and **ResNet** are important CNN architectures, but ResNet is substantially deeper and uses residual connections to make deep networks easier to train.

Comparison AlexNet architecture

Aspect	AlexNet	ResNet
Introduced	2012	2015
Depth	Relatively shallow	Can be very deep, e.g., ResNet-50/101/152
Basic organization	Convolution + pooling + fully connected	Convolution + residual blocks
Filters	Uses convolutional filters of different sizes	Primarily uses smaller convolutional filters in residual blocks
Parameters	About 60 million	Depends on version; ResNet-50 has about 25 million
Feature extraction	Good hierarchical features	More powerful deep hierarchical features
Training difficulty	Easier because it is shallower	Deep networks are difficult without residual connections
Vanishing gradient	Less severe due to shallower depth	Residual connections greatly reduce the problem
Computational cost	Lower than very deep ResNets	Higher for deep variants
Performance	Strong for its time	Generally much better on complex modern recognition tasks

A simplified structure is:

Input image → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Fully Connected → Output

AlexNet was historically important because it demonstrated the effectiveness of deep CNNs for large-scale image recognition.

It introduced important techniques such as:

- ReLU activation
- Dropout
- GPU-based training
- Data augmentation

ResNet architecture

ResNet introduced the **residual block**.

Instead of forcing a layer to learn a complete transformation:

$$\text{Output} = F(x)$$

it learns a residual:

$$\text{Output} = F(x) + x$$

The original input x is passed through a shortcut connection and added to the learned transformation.

Q5. Regularized CNN vs Non-Regularized CNN

The agricultural CNN has **very high training accuracy but poor performance on unseen leaf images**. This is a classic indication of **overfitting**.

Comparison

Aspect	CNN without regularization	CNN with Dropout + Data Augmentation + Batch Normalization
Overfitting	High	Reduced
Generalization	Poor	Better
Training stability	Can be unstable	Generally more stable
Training data requirement	More sensitive to limited data	Data augmentation effectively increases variety
Memory/computation	Lower during training	Some additional computation
Performance on unseen images	Often poor	Usually improved
Robustness	Lower	Higher

1. Dropout

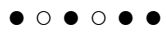
Dropout randomly disables some neurons during training.

For example:

Before dropout:

● ● ● ● ● ●

During training:



The network cannot depend heavily on a small group of neurons.

Benefits

- Reduces overfitting
- Encourages distributed feature learning
- Improves generalization

Dropout is particularly useful in the **fully connected layers**.